

UNIT 3

DYNAMICS



DYNAMICS:

"The part of mechanics that deals with the study of cause of motion is called dynamics."

NEWTON'S FIRST LAW OF MOTION OR LAW OF INERTIA:**Statement:**

"Everybody continues its rest or of uniform motion in a straight line unless it is acted upon by an external unbalanced force."

Explanation:

From our daily life experience, we know that if a body is moved it comes to rest after covering some distance.

Newton's first law predicts that the body must continue its motion along a straight line, but our observation seems to contradict Newton's law.

Actually the body comes to rest because of force of friction and air resistance etc. If there was no force of friction or air resistance, then the body would have continued its motion along straight path with a constant velocity.

First law of Newton's also known as law of INERTIA.**Example:**

We know that book lying on a table remains at rest until someone picks it up (by applying some force). This is in accordance of Newton's first law.

FORCE:

"Force is that agent which changes or tends to change the state of rest or of uniform motion of a body."

$$\text{Net force} = \text{Mass} \times \text{Acceleration}$$

$$\Sigma F = ma$$

Unit:

S.I or M.K.S Unit of force is Newton (N).

INERTIA:

"The property of all material objects to oppose any change in their state of rest or of uniform motion is known as inertia."

NEWTON'S SECOND LAW OF MOTION:**Statement:**

"When an unbalanced force acts on an object it produces acceleration in its own direction which is directly proportional to the magnitude of the force and inversely proportional to the mass of the object."

Mathematically:

If "F" is the magnitude of the force "a" is the acceleration produced in a body of constant mass in the direction of the applied force, then the relationship between acceleration and force can be mathematically expressed as:

$$a \propto F \text{ ----- (i)}$$

$$a \propto 1/m \text{ ----- (ii)}$$

By combining the above equations, we get:

$$a \propto F/m$$

$$a \propto k \times F/m$$

Where k is known as constant of proportionality and its value is 1.

$$a = F/m$$

$$\Sigma F = ma$$

The direction of 'a' and 'F' is the same.

UNIT OF FORCE:

- The S.I or M.K.S Unit of force is NEWTON (N). $N = \text{kg} \cdot \text{m}/\text{s}^2$
- In C.G.S system Unit of force is DYNE, $\text{DYN} = \text{g} \cdot \text{cm}/\text{s}^2$

NEWTON'S THIRD LAW OF MOTION:

Statement:

"For every action (force) there is an equal and opposite reaction (force)."

Explanation:

Action and reaction are two forces exchanged between two bodies when they interact with each other.

Example:

- Motion of a rocket.
- Kicking of a football.
- Our walking.
- Firing on a bullet, gun, and throwing a shell etc.

LINEAR MOMENTUM:

"The product of mass and velocity of a body is called momentum."

OR

"The quantity of motion present in a body is called momentum."

Formula:

$$\text{Momentum} = \text{mass} \times \text{Velocity}$$

$$p = m v$$

Unit:

In S.I system unit of Momentum is Kilogram meter per second ($\text{Kg} \cdot \text{m}/\text{s}$) or Newton Second ($\text{N} \cdot \text{s}$)

IMPULSE:

"The impulse of force is the product of force and time interval (very short) during which it acts."

- Impulse is a vector quantity and is represented with J .

$$J = \Sigma F \cdot \Delta t$$

$$J = \Delta p$$

LINEAR MOMENTUM AND SECOND LAW OR FORCE IN TERMS OF MOMENTUM:

Consider a body of mass 'm' moving with initial velocity V_i , a net force 'F' acts on the body in the direction of velocity for 't' seconds. Such that the velocity of the body changes and becomes V_f . According to Newton's second law of motion:

$$F = ma$$

$$[\text{Where } a = (V_f - V_i) / \Delta t]$$

$$F = m \times ((V_f - V_i) / \Delta t)$$

$$F = mV_f - mV_i$$

[Where Initial momentum of the body $P_i = mV_i$ and Final momentum of the body $P_f = mV_f$]

$$F = \frac{P_f - P_i}{\Delta t}$$

$$F = \frac{\Delta p}{\Delta t}$$

Hence, Rate of change of momentum is equal to force.

LAW OF CONSERVATION OF MOMENTUM:

Statement:

"If no external force acts, then total momentum of an isolated system remains constant i.e. the total momentum before and after collision remains constant."

$$F = \frac{\Delta p}{\Delta t}$$

If no external force acts then,

$$0 = \frac{\Delta p}{\Delta t}$$

By integrating above equation,

$$P = \text{Constant}$$

For two colliding bodies:

For two interacting bodies A and B of masses m_1 and m_2 moving with velocities u_1 and u_2 respectively. After collision they move with velocities v_1 and v_2 respectively along their original direction.

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

Proof:

Consider an isolated system of two bodies masses m_1 and m_2 moving with velocities u_1 and u_2 respectively. Such that $u_1 > u_2$. After collision they move with velocities v_1 and v_2 respectively along their original direction.

Total momentum of the system before collision = $m_1 u_1 + m_2 u_2$ (i)

Total momentum of the system before collision = $m_1 v_1 + m_2 v_2$ (ii)

When the two bodies collide with each other, they come in contact for a time interval Δt . The average force exerted by the body A on B be F_1 , then according to Newton's third law of motion, the body B will also exert a force ($-F$) on the body A.

$$F_A = -F_B$$

$$\left[\text{Where } F = \frac{\Delta p}{\Delta t} \right]$$

$$\frac{\Delta p_A}{\Delta t} = - \frac{\Delta p_B}{\Delta t}$$

$$\Delta P_A = - \Delta P_B$$

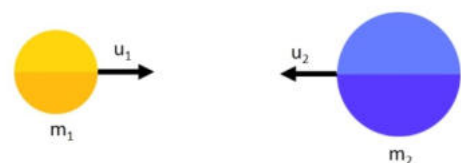
$$(P_f - P_i)_A = - (P_f - P_i)_B$$

[Where P_i = Momentum before collision and P_f = Momentum after collision]

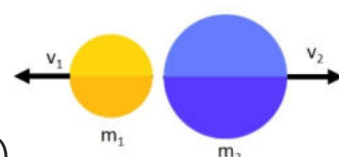
$$(m_1 u_1 - m_1 v_1) = - (m_2 u_2 - m_2 v_2)$$

$$m_1 u_1 - m_1 v_1 = - m_2 u_2 + m_2 v_2$$

Before Collision



After Collision



By rearranging above equation,

$$m_1U_1 + m_2U_2 = m_1V_1 + m_2V_2$$

EXAMPLES OF CONSERVATION OF MOMENTUM:

• Recoil of a gun:

When a gun is fired, the bullet moves with a large velocity and the gun moves with a small velocity in a direction opposite to that of the bullet.

• A man jumps from a boat:

When a man jumps from the boat, the boat slightly moves away from the shore. Initially the total momentum of the boat and the man is zero.

• Rocket propulsion:

The motion of a rocket is based on the principle of conservation of momentum. The rocket and the fuel form an isolated form. Therefore, the total momentum of the system remains constant.

TYPES OF COLLISION:

• Inelastic collision:

"A collision during which the kinetic energy of the system does not remain constant before and after collision but moment remains constant is called inelastic collision."

Examples:

- The collision of rubber ball with hard surface.
- The collision between vehicles.
- Explosion of bombs.

• Elastic collision:

"A collision during which the kinetic energy and the momentum of the system remains constant before and after collision is called elastic collision."

Examples:

- Collision between the molecules of air with the walls of a container at ordinary temperature.
- Compton's scattering of electrons by photons.
- The collision between the atomic and sub atomic particles.

ELASTIC COLLISION IN ONE DIMENSION:

"If the motion of particles remains along the same line before and after collision then the collision is called elastic collision in one dimension."

EXPRESSION FOR FINAL VELOCITIES OF TWO BODIES COLLIDE ELASTICALLY:

Consider two non-rotating spheres of masses m_1 and m_2 moving initially along the line joining their centers with velocities U_1 and U_2 .

U_1 is greater than U_2 so they collide with one another and after having the elastic collision start moving with velocities V_1 and V_2 respectively in the same line and direction.

Momentum of the system:

Momentum of system before collision = $m_1U_1 + m_2U_2$

Momentum of system after collision = $m_1V_1 + m_2V_2$

According to law of conservation of momentum:

Momentum before collision = Momentum after collision

$$\begin{aligned}
 m_1 u_1 + m_2 u_2 &= m_1 v_1 + m_2 v_2 \\
 m_1 u_1 &= m_1 v_1 \\
 m_2 u_2 &= m_2 v_2 \\
 m_1(u_1 - v_1) &= m_2(v_2 - u_2) \dots\dots\dots (1)
 \end{aligned}$$

Kinetic energy of the system:

$$\text{K.E of system before collision} = \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2$$

$$\text{K.E of system after collision} = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

According to law of conservation of energy:

$$\text{K.E of system before collision} = \text{K.E of system after collision}$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$m_1 v_1 - m_1 u_1 = m_2 u_2 - m_2 v_2$$

$$m_1(v_1 - u_1) = m_2(u_2 - v_2) \dots\dots (1)$$

Similarly

$$\text{K.E of the system before collision} = \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2$$

$$\text{K.E of the system after collision} = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

Since the collision is elastic, so the K.E of the system before and after collision is conserved.

Thus

$$\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 = \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2$$

$$\frac{1}{2} (m_1 v_1^2 + m_2 v_2^2) = \frac{1}{2} (m_1 u_1^2 + m_2 u_2^2)$$

$$m_1 v_1^2 - m_1 u_1^2 = m_2 u_2^2 - m_2 v_2^2$$

$$m_1(v_1^2 - u_1^2) = m_2(u_2^2 - v_2^2)$$

$$m_1(v_1 + u_1)(v_1 - u_1) = m_2(u_2 + v_2)(u_2 - v_2) \dots\dots(2)$$

Dividing equation (1) and (2),

Dividing equation (3.10) by equation (3.9)

$$\frac{m_1(v_1 + u_1)(v_1 - u_1)}{m_1(v_1 - u_1)} = \frac{m_2(u_2 + v_2)(u_2 - v_2)}{m_2(u_2 - v_2)}$$

$$v_1 + u_1 = u_2 + v_2$$

From the above equation

$$v_1 = u_2 + v_2 - u_1 \dots\dots(i)$$

$$v_2 = v_1 + u_1 - u_2 \dots\dots(ii)$$

Expression for V_1 Putting the value of v_2 in equation (2)

$$m_1(v_1 - u_1) = m_2(u_2 - v_2)$$

$$m_1(v_1 - u_1) = m_2\{u_2 - (v_1 + u_1 - u_2)\}$$

$$m_1(v_1 - u_1) = m_2\{u_2 - v_1 - u_1 + u_2\}$$

$$m_1(v_1 - u_1) = m_2\{2u_2 - v_1 - u_1\}$$

$$m_1 v_1 - m_1 u_1 = 2m_2 u_2 - m_2 v_1 - m_2 u_1$$

$$m_1 v_1 + m_2 v_1 = m_1 u_1 - m_2 u_1 + 2m_2 u_2$$

$$v_1(m_1 + m_2) = (m_1 - m_2)u_1 + 2m_2 u_2$$

$$V_1 = \frac{(m_1 - m_2) u_1}{(m_1 + m_2)} + \frac{2m_2 u_2}{(m_1 + m_2)}$$

Expression for V_2

In order to obtain V_2 putting the value of V_1 from equation (i) in equation (2)

$$m_1 (v_1 - u_1) = m_2 (u_2 - v_2)$$

$$m_1 (u_2 + v_2 - u_1 - u_1) = m_2 (u_2 - v_2)$$

$$m_1 (u_2 + v_2 - 2u_1) = m_2 (u_2 - v_2)$$

$$m_1 u_2 + m_1 v_2 - 2m_1 u_1 = m_2 u_2 - m_2 v_2$$

$$m_1 v_2 + m_2 v_2 = 2m_1 u_1 + m_2 u_2 - m_1 u_2$$

$$v_2 (m_1 + m_2) = 2m_1 u_1 + (m_2 - m_1) u_2$$

$$V_2 = \frac{2m_1 u_1}{(m_1 + m_2)} + \frac{(m_2 - m_1) u_2}{(m_1 + m_2)}$$

Friction:

"The force which opposes the motion of the body is called friction."

Factor affecting friction:

• Effect of normal reaction:

Force of friction increases with the increase in normal reaction of body which is directly proportional to the weight of the body.

• Nature of surface:

Friction increases with the increasing roughness of surfaces of bodies.

Characteristics of friction:

- The force of friction always acts parallel to the surface in contact and opposite to the direction of motion.
- Although friction is a self-adjusting force, it does not however increase independently with the external force.

TYPES OF FRICTION:

1. STATIC FRICTION:

"When one body is at rest in contact with another, the friction between them is said to be static."

Limiting friction:

"The maximum force of friction between the surfaces of two bodies which are in contact is called limiting friction."

Coefficient of friction:

"The ratio of force of friction to the normal reaction acting between two surfaces in contact is called the coefficient of friction."

Representation:

It is represented by μ .

Mathematically:

$$\mu = \frac{f}{r}$$

Unit and Dimension:

It is ratio between two similar quantities, it has no unit and no dimension.

2. KINETIC OR DYNAMIC FRICTION:

"The friction which exist between two surfaces which are in relative motion is called kinetic or dynamic friction."

Types of kinetic friction:

• Sliding friction:

"When one body is sliding over the other, the friction is said to be sliding."

• Rolling friction:

"When one body is rolls over the other, the friction is said to be rolling."

ANGLE OF FRICTION:

"The angle made by the resultant of the limiting friction and normal reaction with the normal reaction is called angle of friction."

Expression for angle of friction:

From figure, consider ΔOAL ,

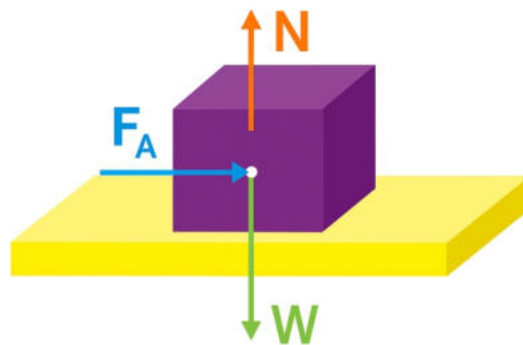
$$\tan(\theta) = \frac{\text{Perp}}{\text{Base}}$$

$$\tan(\theta) = \frac{LA}{OA}$$

$$\tan(\theta) = \frac{f}{R}$$

[where $\frac{f}{R} = \mu$ (Coefficient of friction)]

$$\tan(\theta) = \mu$$



ANGLE

"The maximum angle made by the inclined plane with the horizontal such that the block kept over it does not slip is known as angle of repose."

Expression for angle of repose:

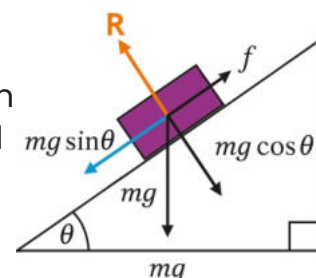
Consider a block of mass 'm' is placed on an inclined plane making an angle 'B' with the horizontal, x-axis is parallel to the inclined plane and y-axis perpendicular to the inclined plane.

The weight $W = mg$ of the block is acting vertically downward.

Components of the weight:

'W' being a vector quantity can be resolved into rectangular components, these are,

- Component parallel to the plane $mg \sin \theta$ opposite to friction 'f'.
- Component perpendicular to the plane $mg \cos \theta$ opposite to Reaction 'R'.



$$mg \sin \theta = -f \dots\dots\dots (1)$$

$$mg \cos \theta = -R \dots\dots\dots (2)$$

Dividing equation (1) and (2),

$$\frac{mg \sin \theta}{mg \cos \theta} = \frac{-f}{-r}$$

$$\frac{\sin \theta}{\cos \theta} = \frac{f}{r}$$

$$\tan \theta = \frac{f}{r}$$

[Where $\frac{f}{r} = \mu$ (Coefficient of friction)]

$$\mu = \tan \theta$$

Angle of repose is equal to the angle of friction.